IBM Model 1

1. **Generative model:**

Break up translation process into smaller steps.
- IBM Model 1 only uses lexical translation.

2. **Translation probability:**

For a foreign sentence, $f = (f_1, \ldots, f_{l_f})$ of length l_f

To a English sentence, $e = (e_1, \ldots, e_{l_e})$ of length l_e

With an alignment of each english word e_j to a foreign word f_i according to the alignment function $a: j \rightarrow$

$$p(e, a|f) = \frac{\epsilon}{(l_f + 1)^{l_e}} \prod_{j=1}^{l_e} t\left(e_j \mid f_{a(j)}\right)$$

where parameter ϵ is a normalization constant.

Example:

Source sentence, f : das Haus ist klein

Target sentence, e : the house is small

das

e	t(e|f)
the	0.7
that	0.15
which	0.075
who	0.05
this	0.025

Haus

e	t(e|f)
house	0.8
building	0.16
home	0.02
household	0.015
shell	0.005

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ist

e	t(e|f)
is	0.8
's	0.16
exists	0.02
has	0.015
are	0.005

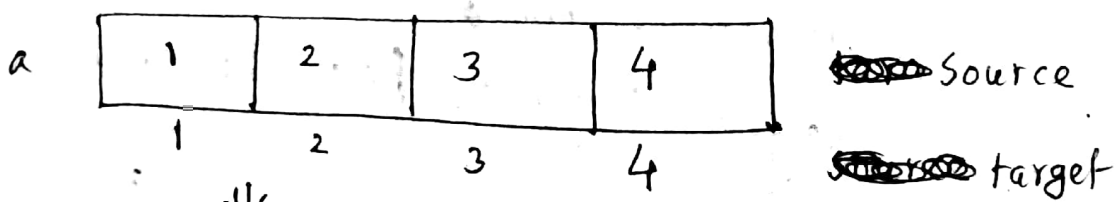
e	t(e|f)
small	0.4
little	0.4
short	0.1
minor	0.06
petty	0.04

Find lexical Translation probability for the given source and target language sentences.

Ans:

$$P(e, a|f) = \frac{\epsilon}{(l_f+1)^{l_e}} \prod_{j=1}^{l_e} t(e_j | f_{a(j)}) \quad \text{..... eqn (1)}$$

	1	2	3	4
Source language sentence, f:	das	Haus	ist	klein
Target language sentence, e:	the	house	is	small
	1	2	3	4



$a: \{1 \to 1, 2 \to 2, 3 \to 3, 4 \to 4\}$

Here, $l_f = 4$

$l_e = 4$

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From eq (1)

$$P(e, a|f) = \frac{\epsilon}{(4+1)^4} \prod_{j=1}^{4} t(e_j | f_{a(j)})$$

$$= \frac{\epsilon}{5^4} \cdot \{ t(e_1 | f_{a(1)}) * t(e_2 | f_{a(2)}) * t(e_3 | f_{a(3)}) * t(e_4 | f_{a(4)}) \}$$

$$= \frac{\epsilon}{5^4} \{ t(e_1 | f_1) * t(e_2 | f_2) * t(e_3 | f_3) * t(e_4 | f_4) \}$$

$$= \frac{\epsilon}{5^4} \{ t(\text{the} | \text{das}) * t(\text{house} | \text{Haus}) * t(\text{is} | \text{ist}) * t(\text{small} | \text{klein}) \}$$

$$= \frac{\epsilon}{5^4} (0.7 * 0.8 * 0.8 * 0.4)$$

$$= \frac{0.7 * 0.8 * 0.8 * 0.4}{625} \quad \text{where } \epsilon = 1$$

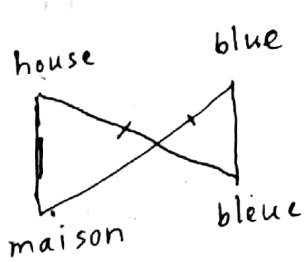
$$= \frac{0.1792}{625}$$

$$= 0.00028672$$

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Expection Maximization algorithm :

To estimate translation values from two sentences.





Example:

maison bleue ⟷ blue house

maison ⟷ house

From the example, maison ⟷ blue
maison ⟷ house
bleue ⟷ blue
bleue ⟷ house

From another, maison → house

step1: set parameter values uniformly

$t(\text{bleue}|\text{house}) = \frac{1}{2}$

$t(\text{maison}|\text{house}) = \frac{1}{2}$

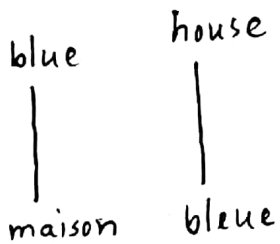
$t(\text{bleue}|\text{blue}) = \frac{1}{2}$

$t(\text{maison}|\text{blue}) = \frac{1}{2}$

Iteration 1:

step2: Compute $P(a, f|e)$ for all alignments

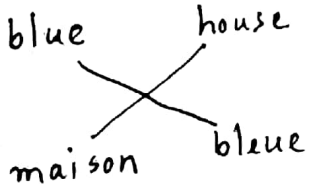
Alignment 1:



$P(a,f|e) = t(\text{maison}|\text{blue}) * t(\text{bleue}|\text{ho}$

$= \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$

Alignment 2:



$P(a,f|e) = t(\text{bleue}|\text{blue}) * t(\text{maison}|\text{house})$

$= \frac{1}{2} * \frac{1}{2} = \frac{1}{4}$

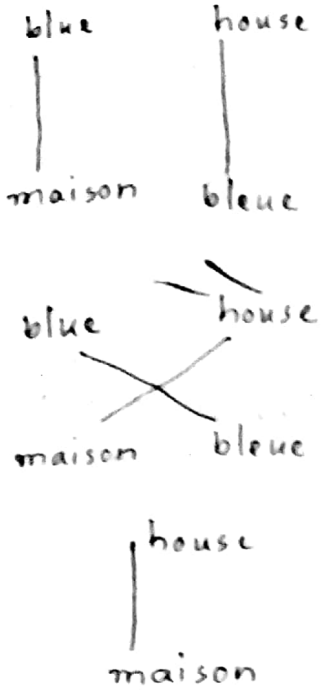
Alignment 3:



$P(a,f|e) = t(\text{maison}|\text{house}) = \frac{1}{2}$

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step3: Normalize $P(a, f|e)$ values to yeild $P(a|e,f)$ values



$$P(a|e,f) = \frac{P(a,f|e)}{\sum_a P(a,f|e)} = \frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{4}} = \frac{\frac{1}{4}}{\frac{2}{4}} = \frac{1}{2}$$

$$P(a|e,f) = \frac{P(a,f|e)}{\sum_a P(a,f|e)} = \frac{\frac{1}{4}}{\frac{1}{4}+\frac{1}{4}} = \frac{1}{2}$$

$$P(a|e,f) = \frac{P(a,f|e)}{\sum_a P(a,f|e)} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$$

[There is only one alignment, so $P(a|e,f)$ will be 1 always]

step4: Collect fractional counts

$tc(\text{maison}|\text{blue}) = \frac{1}{2}$

$tc(\text{bleue}|\text{house}) = \frac{1}{2}$

$tc(\text{bleue}|\text{blue}) = \frac{1}{2}$

$tc(\text{maison}|\text{house}) = \frac{1}{2} + 1 = \frac{3}{2}$

step5: Normalize fractional counts to get revised parameter values

$$t(\text{maison}|\text{blue}) = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{2}} = \frac{1}{2}$$

$$t(\text{bleue}|\text{house}) = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{3}{2}} = \frac{1}{2} * \frac{2}{4} = \frac{1}{4}$$

$$t(\text{bleue}|\text{blue}) = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{2}} = \frac{1}{2}$$

$$t(\text{maison}|\text{house}) = \frac{\frac{3}{2}}{\frac{1}{2}+\frac{3}{2}} = \frac{3}{2} * \frac{2}{4}$$

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Iteration2 :

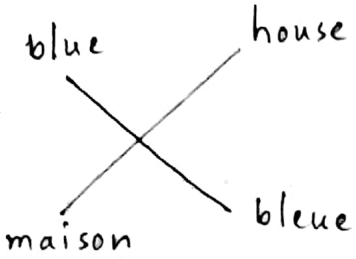
step2 : Compute $P(a, f|e)$ for all alignments

Alignment-I



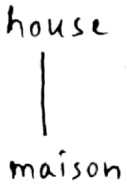
$$P(a, f|e) = t(\text{maison}|\text{blue}) * t(\text{bleue}|\text{house})$$
$$= \frac{1}{2} * \frac{1}{4} = \frac{1}{8}$$

Alignment-II



$$P(a, f|e) = t(\text{bleue}|\text{blue}) * t(\text{maison}|\text{house})$$
$$= \frac{1}{2} * \frac{3}{4} = \frac{3}{8}$$

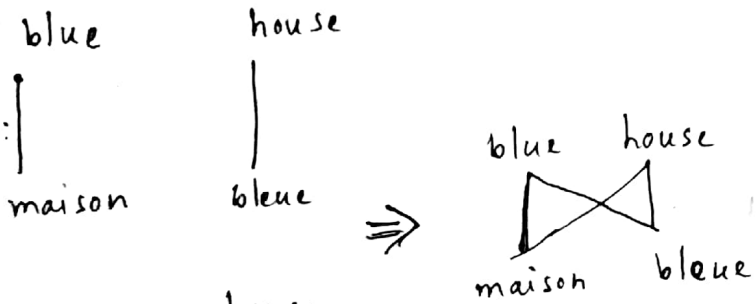
Alignment-III



$$P(a, f|e) = t(\text{maison}|\text{house})$$
$$= \frac{3}{4}$$

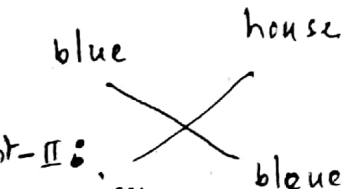
step3 : Normalize $P(a, f|e)$ values to yeild $P(a|e, f)$ values

Alignment-I:



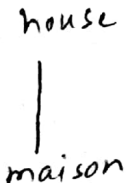
$$P(a|e, f) = \frac{P(a, f|e)}{\sum P(a, f|e)}$$
$$= \frac{\frac{1}{8}}{\frac{1}{8} + \frac{3}{8}}$$
$$= \frac{1}{4}$$

Alignment-II :



$$P(a|e, f) = \frac{P(a, f|e)}{\sum P(a, f|e)}$$
$$= \frac{\frac{3}{8}}{\frac{1}{8} + \frac{3}{8}}$$
$$= \frac{3}{4}$$

Alignment-III :



$$P(a|e, f) = \frac{P(a, f|e)}{\sum P(a, f|e)} = \frac{\frac{3}{4}}{\frac{3}{4}} = 1 = \frac{3}{4}$$

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house)

house)

step4: Collect fractional counts

$tc(\text{maison} \mid \text{blue}) = \frac{1}{4}$

$tc(\text{bleue} \mid \text{house}) = \frac{1}{4}$

$tc(\text{bleue} \mid \text{blue}) = \frac{3}{4}$

$tc(\text{maison} \mid \text{house}) = \frac{3}{4} + 1 = \frac{7}{4}$

step5: Normalize fractional counts to get revised parameter values

$$t(\text{maison} \mid \text{blue}) = \frac{\frac{1}{4}}{\sum tc(\ldots \mid \text{blue})}$$

$$= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{3}{4}} = \frac{1}{4}$$

$$t(\text{bleue} \mid \text{house}) = \frac{\frac{1}{4}}{\sum tc(\ldots \mid \text{house})}$$

$$= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{7}{4}}$$

$$= \frac{1}{8}$$

$$t(\text{bleue} \mid \text{blue}) = \frac{\frac{3}{4}}{\sum tc(\ldots \mid \text{blue})} = \frac{\frac{3}{4}}{\frac{1}{4} + \frac{3}{4}} = \frac{3}{4}$$

$$t(\text{maison} \mid \text{house}) = \frac{\frac{7}{4}}{\frac{1}{4} + \frac{7}{4}} = \frac{\frac{7}{4}}{\frac{8}{4}} = \frac{7}{8}$$

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Repeating steps 2-5 many times yields

$t(\text{bleue} \mid \text{house}) = 0.0001$

$t(\text{maison} \mid \text{house}) = 0.9999$ ✓

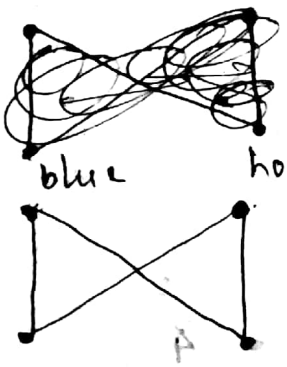
$t(\text{bleue} \mid \text{blue}) = 0.9999$ ✓

$t(\text{maison} \mid \text{blue}) = 0.0001$

maison bleue ⟷ blue house

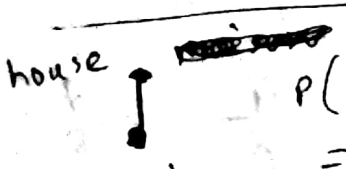
maison ⟷ house

So we can calculate translation probabilites by the way of these alignment probabilites



$$P(\underline{f} \mid \underline{e}) = \sum_{a} P(a, f \mid e)$$

$$= \sum_{a} \prod_{j=1}^{m} t(f_j \mid e_i)$$



$P(f \mid e)$

$= \sum_{a} P(a, f \mid e)$

$= t(\text{maison} \mid \text{blue}) * t(\text{blue} \mid \text{house})$

$+ t(\text{bleue} \mid \text{blue}) * t(\text{maison}$

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